# Isolation $E_T$ Definition Comparison: 60 MeV Tower Threshold vs. 120 MeV vs. Topo-Cluster $R = 0.4$

PPG12 Automated Investigation

April 24, 2026

## 1 Motivation

The isolated photon cross-section measurement relies on a reconstruction-level isolation energy cut to define the signal region in the ABCD method. Three isolation definitions are available in the slimtree:

- **60 MeV tower threshold**: Sum tower $E_T$ in $\Delta R < 0.3$ cone (EMCal + IHCal + OHCal), including only towers above 60 MeV. Branches: `cluster_iso_03_60_{emcal,hcalin,hcalout}`.
- **120 MeV tower threshold**: Same cone and calorimeters, but with a 120 MeV tower threshold. Branches: `cluster_iso_03_120_{emcal,hcalin,hcalout}`.
- **Topo-cluster** $R = 0.4$: Sum $E_T$ of topological clusters in $\Delta R < 0.4$ cone. Branch: `cluster_iso_topo_04`. This is the **current nominal** definition (`use_topo_iso: 2` in the config).

This note quantifies the differences in run-by-run stability, ABCD yield, and signal/background discrimination power.

## 2 Signal/Background Discrimination (ROC Curves)

Using truth-matched signal (direct + fragmentation photons with truth $E_T^{\text{iso}} < 4\,\text{GeV}$ in $R = 0.3$) and background (jet-sample clusters) from the full MC simulation chain, we compute ROC curves for all seven available isolation definitions.

Table 1 summarizes the AUC values for the three primary definitions and all seven available.

Table 1: ROC AUC values by isolation definition and $E_T$ bin.

| Definition | 10–15 GeV | 15–20 GeV | 20–25 GeV | 25–30 GeV |
|---|---|---|---|---|
| 70 MeV tower | 0.820 | 0.849 | 0.857 | 0.861 |
| **60 MeV tower** | **0.806** | **0.847** | **0.855** | **0.859** |
| **120 MeV tower** | **0.805** | **0.847** | **0.857** | **0.857** |
| topo $R = 0.3$ | 0.804 | 0.842 | 0.853 | 0.858 |
| **topo** $R = 0.4$ | **0.823** | **0.871** | **0.875** | **0.878** |
| topo soft $R = 0.3$ | 0.805 | 0.844 | 0.857 | 0.862 |
| topo soft $R = 0.4$ | 0.827 | 0.873 | 0.881 | 0.883 |

**Key findings:**

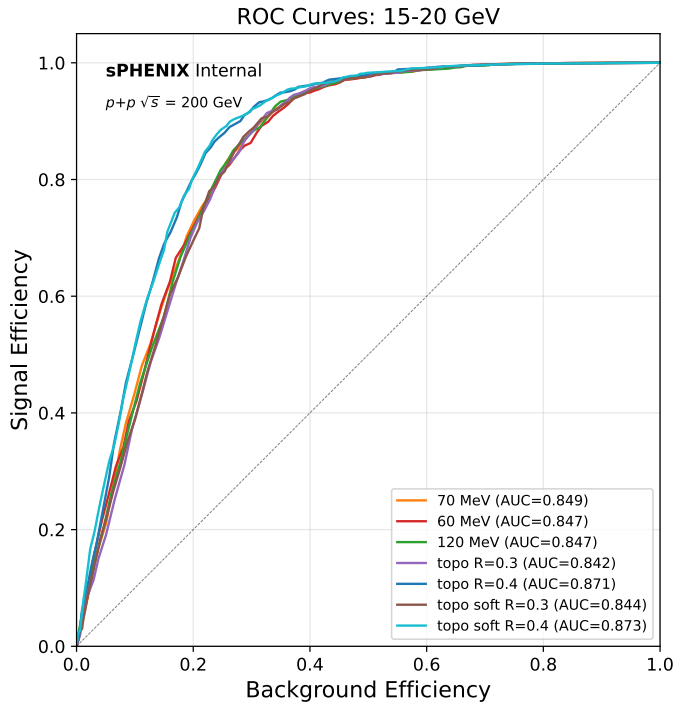


Figure 1: ROC curves for the 15–20 GeV $E_T$ bin. The topo $R = 0.4$ definitions (blue/cyan) are clearly separated from the tower-threshold methods. AUC values are shown in the legend.

- Topo $R = 0.4$ outperforms tower-threshold methods by $\sim$2% in AUC across all $E_T$ bins. The advantage is largest at low $E_T$ (10–15 GeV: 0.823 vs. 0.806).
- 60 MeV and 120 MeV tower thresholds are essentially identical (<0.3% AUC difference). The choice of tower threshold is negligible for discrimination.
- **The dominant factor is cone size, not method.** All $R = 0.3$ methods (tower 60/70/120 MeV, topo $R = 0.3$, topo soft $R = 0.3$) cluster together at AUC $\approx$ 0.80–0.86, while both $R = 0.4$ methods stand $\approx$2% above (Figure 3). The topo clustering algorithm itself provides minimal gain at fixed cone size; the larger cone captures more jet activity around background clusters, driving the improved rejection.
- Topo soft $R = 0.4$ is marginally better than standard topo $R = 0.4$ (+0.2–0.5% AUC).
- Among tower-threshold methods at $R = 0.3$, the 70 MeV threshold slightly outperforms 60 MeV at low $E_T$ (0.82 vs. 0.81) but they converge at high $E_T$. 120 MeV is indistinguishable from 60 MeV.

## 3 Isolation $E_T$ Distributions

The distribution shapes (Figure 4) explain the discrimination difference:

- **Signal**: Topo $R = 0.4$ has the sharpest peak near $E_T^{\text{iso}} = 0$, concentrating more signal at low isolation values.

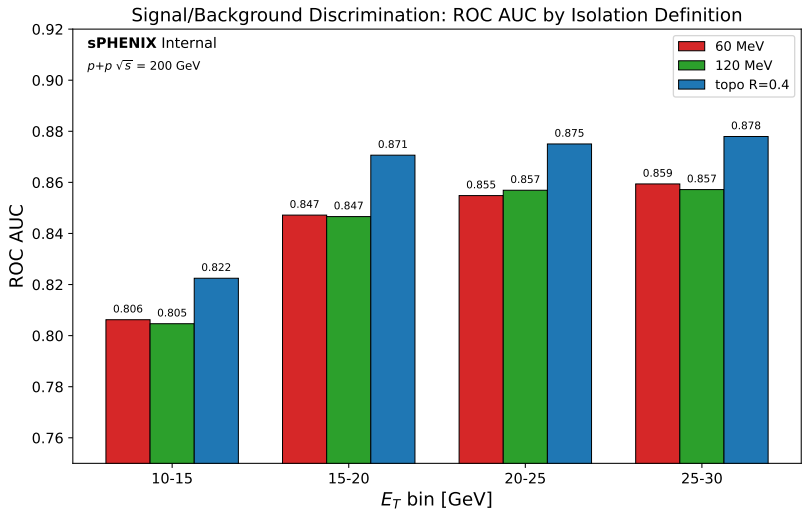


Figure 2: ROC AUC comparison across $E_T$ bins for the three definitions of interest. Topo $R = 0.4$ consistently outperforms both tower-threshold methods by $\approx 2\%$ in AUC.

- **Background**: Tower-threshold methods produce smooth, broad distributions. Topo $R = 0.4$ has a distinctive spike near zero followed by a broader tail, yielding better separation.
- 120 MeV shifts both signal and background to lower $E_T^{\text{iso}}$ compared to 60 MeV (fewer towers contribute), but the shift is similar for both categories, so discrimination is unchanged.

# 4 Run-by-Run Stability

Using the existing run-by-run QA infrastructure (`Cluster_rbr.C`), we compare the mean isolation $E_T$ and ABCD region yields per run for all three definitions across all 1428 analyzed runs.

Table 2: Run-by-run stability summary (1428 runs).

| Definition | $\langle E_T^{\text{iso}} \rangle$ [GeV] | RMS/Mean | Yield $A$ [mb$^{-1}$] | RMS/Mean $A$ | Yield $C$ [mb$^{-1}$] |
|---|---|---|---|---|---|
| 60 MeV tower | 2.088 | 6.5% | 6963 | 19.4% | 10455 |
| 120 MeV tower | 1.375 | 6.4% | 10702 | 17.0% | 17990 |
| topo $R = 0.4$ | 1.993 | 4.6% | 8606 | 17.4% | 12148 |

**Stability findings:**

- Topo $R = 0.4$ has **30% less run-by-run variation** in mean $E_T^{\text{iso}}$ (RMS/Mean 4.6% vs. 6.4–6.5% for tower methods).
- 120 MeV has a much lower mean isoET (1.38 vs. 2.09 GeV for 60 MeV) because the higher threshold removes towers between 60–120 MeV from the isolation sum.
- 120 MeV yields are **artificially inflated** (Yield $A = 10702$ vs. 6963 for 60 MeV) because the same parametric cut $E_T^{\text{iso}} < b + s \cdot E_T$ was optimized for topo $R = 0.4$, not re-tuned for the lower isoET scale. This does not reflect better physics performance.
- All three definitions show the same crossing-angle structure and outlier runs.

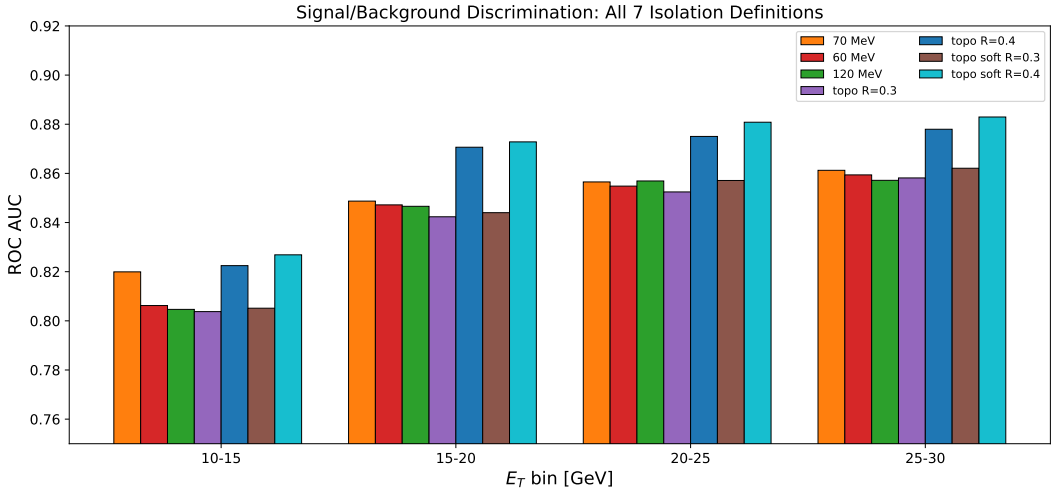


Figure 3: ROC AUC for all 7 isolation definitions. The hierarchy is dominated by cone size: all $R = 0.3$ methods cluster together, while $R = 0.4$ methods stand above.

### 4.1 ABCD Ratio Stability

The $C/A$ ratio (Figure 7) is the key ABCD closure metric. All three definitions produce consistent ratios ($\sim$1.5 in 0 mrad, $\sim$1.2 in 1.5 mrad), with the same outlier runs. This means the **ABCD background subtraction is insensitive to the isolation definition choice**.

## 5 Summary and Recommendation

Table 3: Overall comparison of the three isolation definitions.

| Metric | 60 MeV | 120 MeV | topo $R = 0.4$ |
|---|---|---|---|
| ROC AUC (15–20 GeV) | 0.847 | 0.847 | **0.871** |
| Mean isoET RMS/Mean | 6.5% | 6.4% | **4.6%** |
| Region A yield* | 6963 | 10702 | **8606** |
| C/A ratio stability | OK | OK | **OK** |
| Available in sim? | Yes | Yes | Partial |

*Region A yields are not directly comparable across definitions because the same parametric cut was applied without re-optimization. The 120 MeV yield is inflated by the lower isoET scale.

**Bottom line:**

1. **Topo $R = 0.4$ is the best choice** for the nominal analysis: best discrimination (+2% AUC), best run-by-run stability (30% less variation), and a favorable yield/purity balance.

2. **60 MeV vs. 120 MeV is a wash**: <0.3% difference in discrimination, essentially identical performance. The tower threshold choice has negligible physics impact.

3. **Cone size $R = 0.4$ is the key driver**, not the topo algorithm itself. All $R = 0.3$ methods (tower and topo) cluster together in AUC.

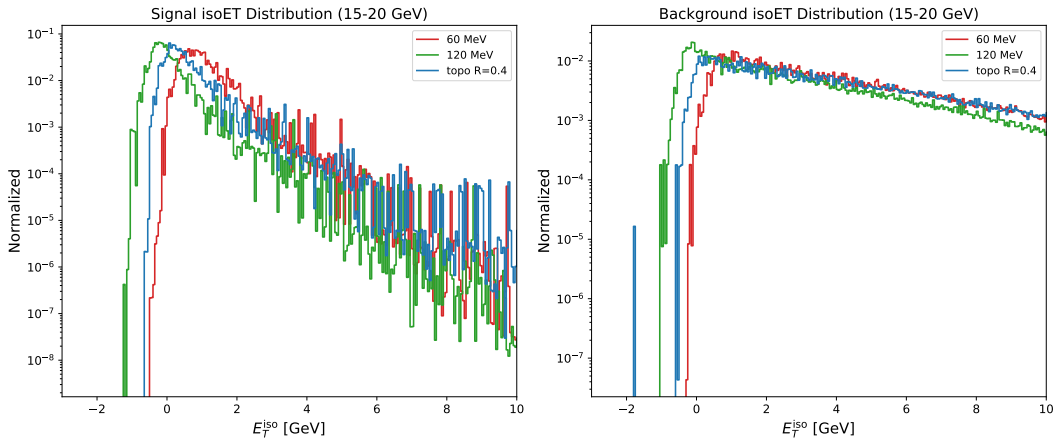


Figure 4: Normalized isolation $E_T$ distributions for signal (left) and background (right) in the 15–20 GeV bin. Topo $R = 0.4$ produces a sharper signal peak near zero and a more spread-out background tail.

4. The current nominal config (`use_topo_iso: 2`) is **well justified**.

5. The 60 MeV tower threshold remains a valid cross-check definition for systematic studies (available in both data and sim).

# 6 Caveats (from Physics Review)

1. **Cone size conflation (WARN):** The topo $R = 0.4$ comparison mixes two effects: topo-cluster algorithm and larger cone size. The ROC analysis includes all 7 definitions (including topo $R = 0.3$) and shows the dominant factor is $R = 0.4$ vs. $R = 0.3$, not tower vs. topo. The run-by-run QA only compares three definitions and conflates these.

2. **Parametric cut not re-optimized (WARN):** The same $E_T^{\text{iso}} < b + s \cdot E_T$ cut (optimized for topo $R = 0.4$) is applied to all three definitions in the run-by-run ABCD. This makes yield comparisons unfair (especially 120 MeV, whose lower isoET scale inflates yields). The ROC comparison is unaffected (no cut applied).

3. **Missing intermediate jet samples (WARN):** The `IsoROC_calculator.C` uses jet5/12/20/30/40 but not jet8/10/15. The jet5 window (7–14 GeV) overextends into the jet8 domain, causing slight background misweighting at low $E_T$. Effect on ROC AUC is modest since curves are normalized.

4. **Config file (WARN):** `config_bdt_isoroc.yaml` does not exist; IsoROC was run with a manually provided config.

# 7 Code Changes

- `efficiencytool/Cluster_rbr.C`: Removed hardcoded `use_topo_iso = 0` override; added `iso_threshold == 2` for 120 MeV support; output filename now includes `var_type`.

- `efficiencytool/config_bdt_rbr_120MeV.yaml`: New config for 120 MeV run-by-run QA.

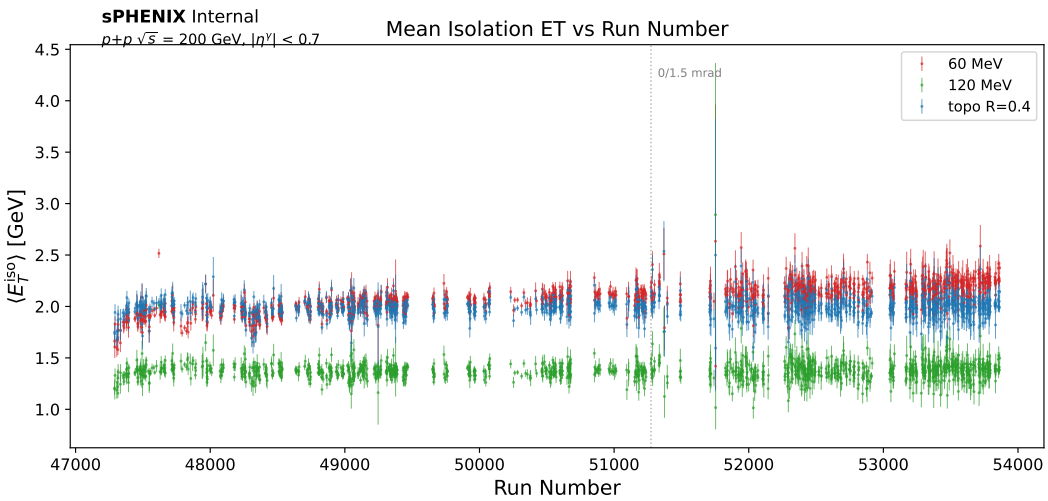


Figure 5: Mean isolation $E_T$ vs. run number. Topo $R = 0.4$ (blue) runs $\sim$0.1 GeV lower than 60 MeV (red) with visibly less scatter. The dashed line marks the 0/1.5 mrad crossing angle boundary.

- `plotting/plot_isoET_comparison.py`: New comparison plotting script (reads rbrQA files + ROC merged file, produces overlay plots and summary table).

# 8 Files Produced

- `figures/isoET_comparison/*.pdf` — comparison plots
- `efficiencytool/roc_plots/*.pdf` — per-$E_T$-bin ROC curves
- `reports/isolation_ET_comparison.pdf` — this report

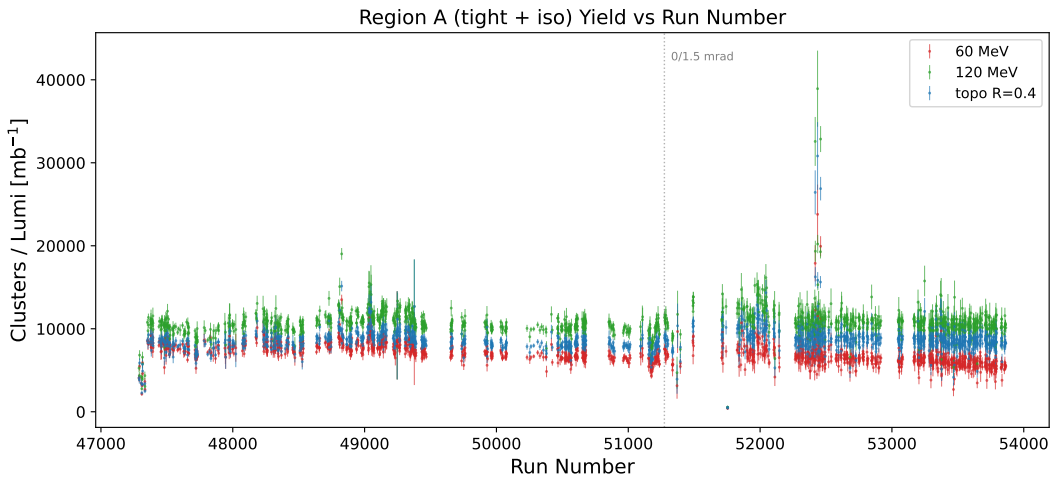


Figure 6: Region A (tight + isolated) cluster yield per lumi vs. run number. Topo $R = 0.4$ yields ≈24% more signal-region clusters, consistent with the narrower isolation distribution.

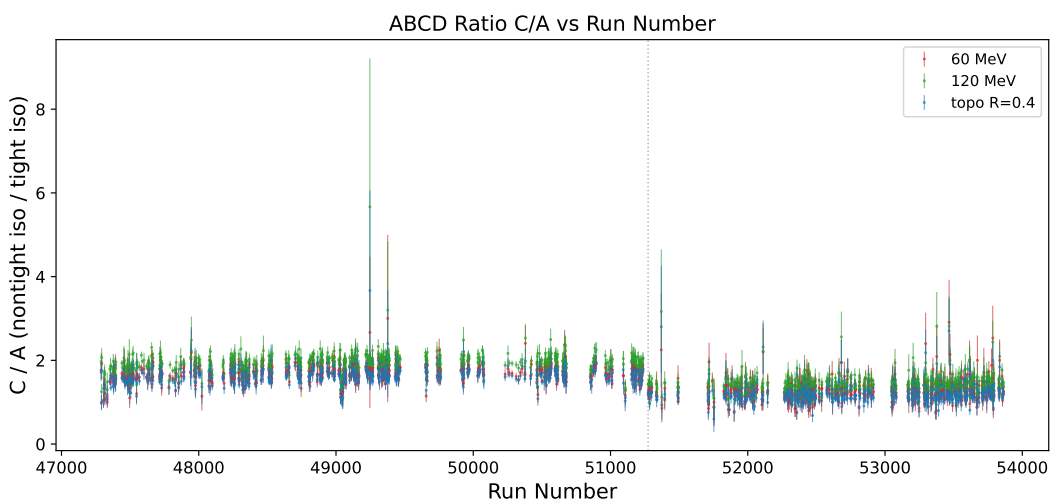


Figure 7: Ratio $C/A$ (nontight-iso / tight-iso) vs. run number. All three definitions produce similar ratios, confirming that the ABCD closure is robust against the isolation definition choice.